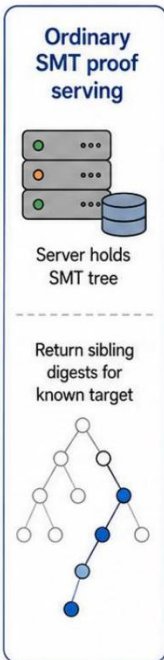


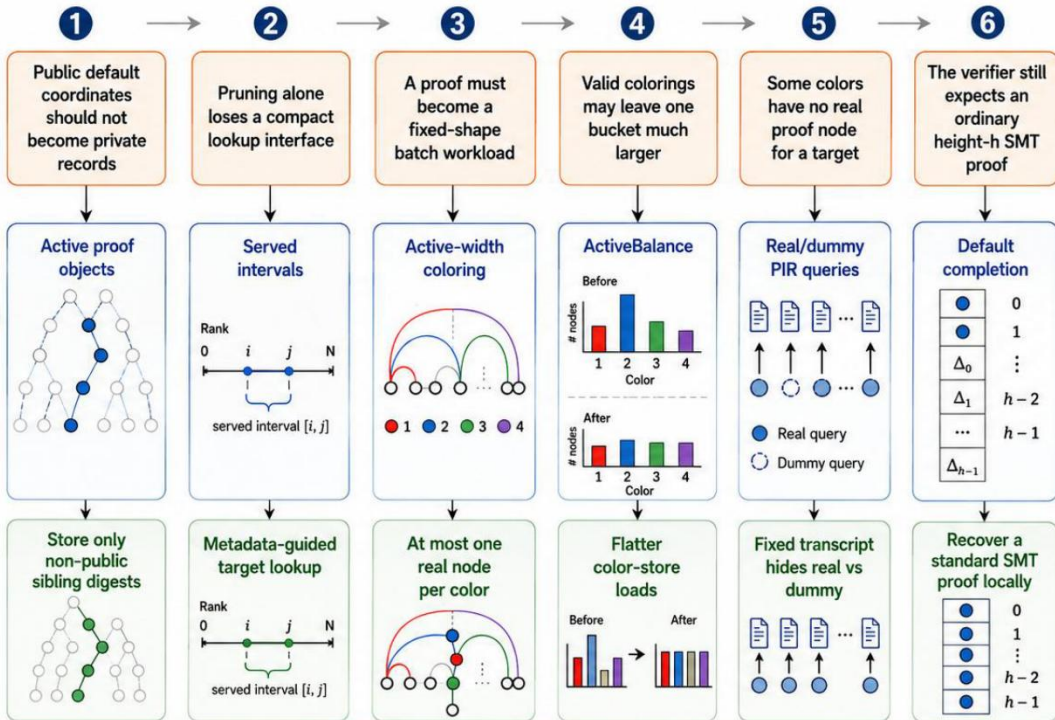
# SparseTreePIR: Solving the SMT-to-Private-Retrieval Challenge



Challenge

SparseTreePIR mechanism

Result

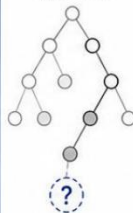


Target-private SMT proof retrieval



Client receives completed proof

Target remains hidden from the server



Private records: active proof-bearing nodes



Batch width: active width  $m$



Proof semantics unchanged: ordinary height- $h$  SMT proof